

Master's Thesis

A Comparison of Base Forecasting Approaches for Spatio-Temporal Reconciliation

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Abstract

[TBD: Context, research question, methods, key results after $h=1$ full run.]

Contents

Abstract	I
List of Abbreviations	IV
List of Symbols	V
List of Figures	V
List of Tables	VI
1 Introduction	1
1.1 Research Question	1
1.2 Contributions	1
1.3 Thesis Outline	1
2 Theoretical Background	2
2.1 Hierarchical Time Series	2
2.2 Forecast Reconciliation	2
2.3 Base Forecasting Approaches	2
2.4 Related Work	2
3 Data and Experimental Design	3
3.1 Dataset	3
3.2 Spatial Hierarchy	3
3.3 Temporal Aggregation	3
3.4 Rolling-Origin Evaluation	3
3.5 Reconciliation Pipeline	4
3.6 Evaluation Metrics	4
4 Base Forecasting Methods	5
4.1 Pipeline Overview	5
4.2 SARIMAX	5
4.3 SARIMAX with NWP Replacement	5
4.4 Random Forest	5
4.5 Random Forest with NWP Replacement	5
4.6 LightGBM	5
4.7 LightGBM with NWP Replacement	5

4.8	ETS (Author Implementation)	6
4.9	ETS (This Implementation)	6
5	Results and Discussion	7
5.1	Overview	7
5.2	Results Tables	7
5.3	Discussion	7
5.4	Limitations	9
6	Conclusion	10
6.1	Summary	10
6.2	Outlook	10
	Bibliography	11
	Appendices	12
A	Reconciliation Implementation	12
B	Replication Bug Fixes	12
C	Supplementary Results	12

List of Abbreviations

NWP	Numerical Weather Prediction
PV	Photovoltaic
SARIMAX	Seasonal Autoregressive Integrated Moving Average with eXogenous variables
RF	Random Forest
LGBM	Light Gradient Boosting Machine
ETS	Exponential Smoothing State Space model
CTWLSV	Cross-Temporal Weighted Least Squares Variances reconciliation
CTBU	Cross-Temporal Bottom-Up reconciliation
PERS	Persistence baseline forecast
nRMSE	normalized Root Mean Square Error
nMBE	normalized Mean Bias Error
SMAPE	Symmetric Mean Absolute Percentage Error

List of Figures

1	Spatial hierarchy of the PV forecasting system.	3
2	Temporal aggregation levels from hourly ($k = 1$) to daily ($k = 24$).	3
3	Rolling-origin evaluation scheme.	4
4	Forecasting pipeline from base methods to evaluation.	5

List of Tables

1	Mean nRMSE (%) by base method, reconciliation approach, spatial level, and temporal resolution.	7
2	Mean nMBE (%) by base method, reconciliation approach, spatial level, and temporal resolution.	8
3	Mean Skill score (%) relative to PERS by base method, reconciliation approach, spatial level, and temporal resolution.	8
4	Mean SMAPE (%) by base method, reconciliation approach, spatial level, and temporal resolution.	9

1 Introduction

[TBD: Motivation for solar PV forecasting and spatio-temporal reconciliation.]

1.1 Research Question

This thesis addresses the following central research question:

How do different base forecasting approaches compare when applied within the spatio-temporal reconciliation framework of Di Fonzo and Girolimetto [2023]?

While Di Fonzo and Girolimetto [2023] fixed the base forecast (ETS with NWP) and compared reconciliation methods, this work reverses that design: we fix the two best-performing reconciliation methods from their study (CTWLSV and CTBU) and systematically compare eight base forecasting variants across four model families.

1.2 Contributions

[TBD after full experimental run. Planned contributions:]

- A systematic comparison of statistical and machine-learning base forecasts under a common spatio-temporal hierarchy.
- An assessment of NWP integration strategies (integrated vs. L2 hourly replacement) across model families.
- A reproducible implementation extending the experimental setup of Di Fonzo and Girolimetto [2023].

1.3 Thesis Outline

Chapter 2 reviews hierarchical forecasting and reconciliation theory. Chapter 3 describes the dataset, spatial and temporal hierarchy, and evaluation design. Chapter 4 specifies the eight base forecasting methods. Chapter 5 presents and discusses the results. Chapter 6 concludes with limitations and outlook.

2 Theoretical Background

2.1 Hierarchical Time Series

[TBD: Spatial hierarchy L_0 , L_1 , L_2 ; temporal aggregation levels. Cite Hyndman and Athanasopoulos [2021], Wickramasuriya et al. [2019].]

2.2 Forecast Reconciliation

[TBD: Coherence constraints, optimal reconciliation. Brief treatment—reconciliation methods are reproduced from Di Fonzo and Girolimetto [2023], not compared in this thesis.]

2.3 Base Forecasting Approaches

[TBD: Overview of SARIMAX, Random Forest, LightGBM, and ETS. Cite Breiman [2001], Ke et al. [2017], Hyndman et al. [2011].]

2.4 Related Work

[TBD: Yang et al. sequential reconciliation; Di Fonzo & Girolimetto spatio-temporal extension [Di Fonzo and Girolimetto, 2023].]

3 Data and Experimental Design

3.1 Dataset

[TBD: `meas.csv` (318 PV stations, hourly, 365 days), `pred.csv` (NWP predictions).]

3.2 Spatial Hierarchy

[TBD: L_0 Total (1), L_1 Regions (5), L_2 Stations (318). See Figure 1.]

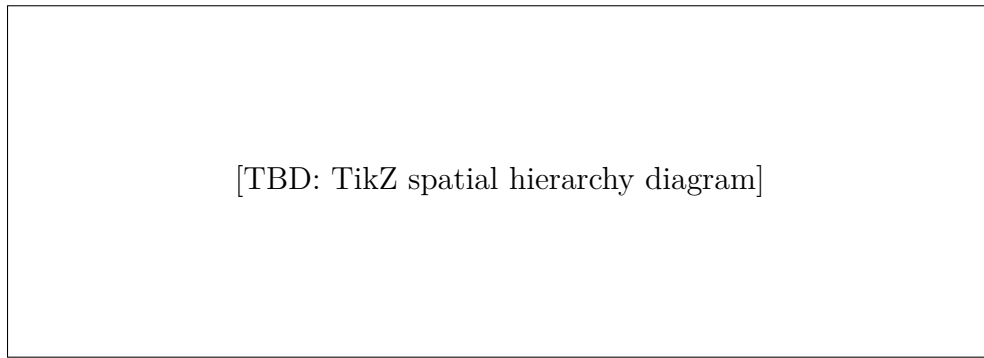


Figure 1: Spatial hierarchy of the PV forecasting system.

3.3 Temporal Aggregation

[TBD: Temporal levels $k \in \{1, 2, 3, 4, 6, 8, 12, 24\}$, $k^* = 60$. See Figure 2.]

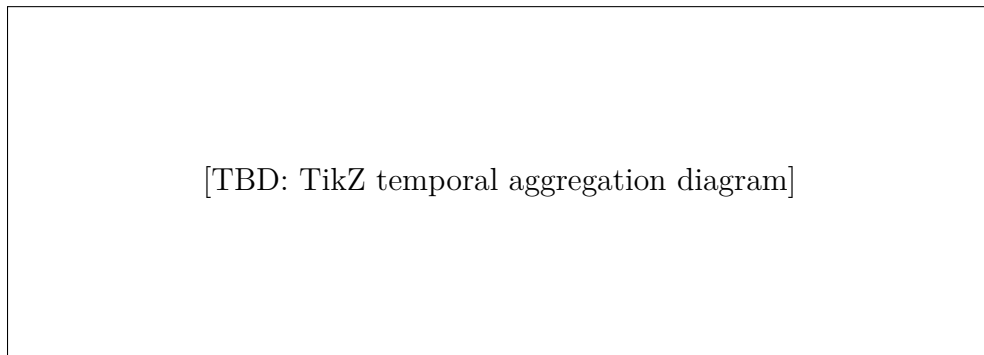


Figure 2: Temporal aggregation levels from hourly ($k = 1$) to daily ($k = 24$).

3.4 Rolling-Origin Evaluation

Following Di Fonzo and Girolimetto [2023], we use a rolling-origin design with a 14-day training window (336 hours), a forecast horizon of two days (48 hourly forecasts), and 350 replications. As in the reference paper, forecast evaluation considers only the day-2

(*operating day*) forecasts, consistent with operational forecast submission requirements. See Figure 3.

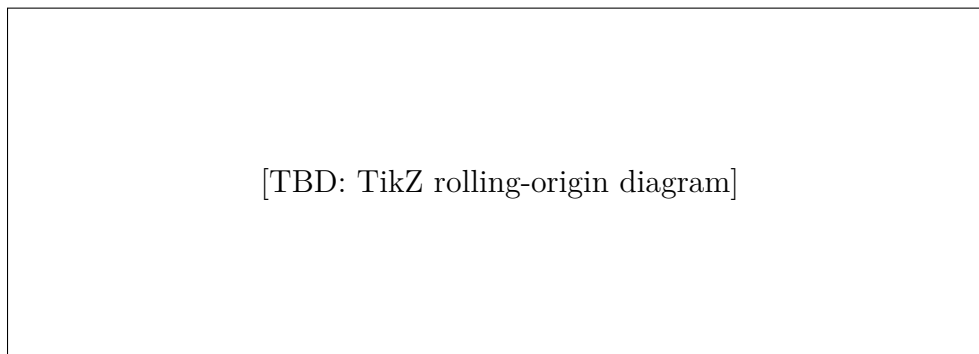


Figure 3: Rolling-origin evaluation scheme.

3.5 Reconciliation Pipeline

[TBD: Two fixed reconciliation methods—CTWLSV (`FoReco::octrec`) and CTBU (`FoReco::thfrec`)—reproduced from Di Fonzo and Girolimetto [2023].]

3.6 Evaluation Metrics

[TBD: nRMSE, nMBE, Skill (vs. PERS), SMAPE. Formulas with nomenclature symbols.]

4 Base Forecasting Methods

4.1 Pipeline Overview

[TBD: Base forecast generation → NWP post-processing (for *_nwp variants) → CTWLSV/CTBU reconciliation → evaluation. See Figure 4.]

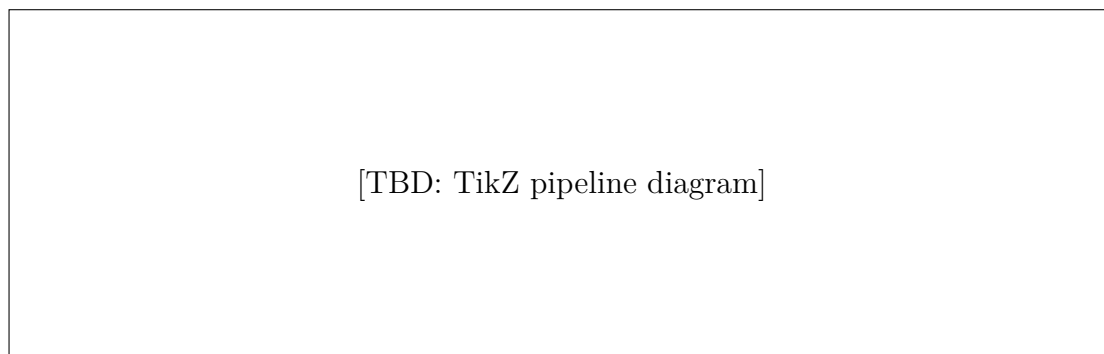


Figure 4: Forecasting pipeline from base methods to evaluation.

4.2 SARIMAX

[TBD: `auto.arima()` with NWP as exogenous regressor.]

4.3 SARIMAX with NWP Replacement

[TBD: L2 hourly ($k = 1$) replaced by NWP via post-processing.]

4.4 Random Forest

[TBD: 500 trees, 12 features, recursive forecasting.]

4.5 Random Forest with NWP Replacement

[TBD]

4.6 LightGBM

[TBD: 300 rounds, same features as RF.]

4.7 LightGBM with NWP Replacement

[TBD]

4.8 ETS (Author Implementation)

[TBD: `thief::th.forecast()` per temporal level, original author code from Di Fonzo and Girolimetto [2023].]

4.9 ETS (This Implementation)

[TBD: Per-level ETS with NWP hybrid at L2 hourly level.]

5 Results and Discussion

5.1 Overview

[TBD: Evaluation at L_0 , L_1 , L_2 spatial levels and hourly ($k = 1$) and daily ($k = 24$) temporal resolutions. Each table contains 17 rows: PERS baseline plus eight base methods reconciled via CTWLSV and CTBU.]

5.2 Results Tables

Table 1: Mean nRMSE (%) by base method, reconciliation approach, spatial level, and temporal resolution.

Base method	Reconciliation	L0 H	L0 D	L1 H	L1 D	L2 H	L2 D
PERS	—	34.60	20.22	43.13	24.56	59.74	30.65
SARIMAX	CTWLSV	27.79	10.95	33.93	15.00	50.59	21.92
SARIMAX	CTBU	28.41	11.00	34.45	15.02	50.97	21.94
SARIMAX+NWP	CTWLSV	22.58	10.39	30.26	14.61	48.16	21.62
SARIMAX+NWP	CTBU	22.79	10.38	30.48	14.58	48.29	21.60
RF	CTWLSV	24.10	13.58	31.12	17.23	45.16	22.58
RF	CTBU	24.16	13.60	31.16	17.25	45.19	22.60
RF+NWP	CTWLSV	23.78	13.39	30.84	17.03	45.46	22.28
RF+NWP	CTBU	24.15	13.41	31.22	17.04	45.74	22.29
LightGBM	CTWLSV	22.89	12.57	30.34	16.46	47.57	23.13
LightGBM	CTBU	22.99	12.61	30.41	16.48	47.61	23.14
LightGBM+NWP	CTWLSV	23.14	12.71	30.58	16.49	48.03	22.87
LightGBM+NWP	CTBU	23.91	12.83	31.24	16.58	48.44	22.92
ETS (author)	CTWLSV	22.86	13.27	29.53	16.72	43.78	21.98
ETS (author)	CTBU	22.78	13.11	29.50	16.55	43.80	21.88
ETS	CTWLSV	22.83	13.15	29.47	16.56	43.66	21.71
ETS	CTBU	22.74	12.97	29.43	16.37	43.67	21.58

5.3 Discussion

[TBD after h=1 full run:]

- Overall ranking across spatial and temporal levels
- Effect of NWP variants within each model family
- Sensitivity to reconciliation method (CTWLSV vs. CTBU)
- Comparison to the ETS+NWP baseline of Di Fonzo and Girolimetto [2023]

Table 2: Mean nMBE (%) by base method, reconciliation approach, spatial level, and temporal resolution.

Base method	Reconciliation	L0 H	L0 D	L1 H	L1 D	L2 H	L2 D
PERS	—	-0.10	-0.10	-0.09	-0.09	-0.10	-0.10
SARIMAX	CTWLSV	-3.24	-3.24	-3.03	-3.03	-3.32	-3.32
SARIMAX	CTBU	-3.35	-3.35	-3.14	-3.14	-3.43	-3.43
SARIMAX+NWP	CTWLSV	-1.51	-1.51	-1.48	-1.48	-1.54	-1.54
SARIMAX+NWP	CTBU	-1.38	-1.38	-1.32	-1.32	-1.41	-1.41
RF	CTWLSV	-2.36	-2.36	-2.29	-2.29	-2.38	-2.38
RF	CTBU	-2.41	-2.41	-2.34	-2.34	-2.43	-2.43
RF+NWP	CTWLSV	-1.71	-1.71	-1.67	-1.67	-1.73	-1.73
RF+NWP	CTBU	-1.73	-1.73	-1.68	-1.68	-1.74	-1.74
LightGBM	CTWLSV	-2.14	-2.14	-2.08	-2.08	-2.16	-2.16
LightGBM	CTBU	-2.18	-2.18	-2.12	-2.12	-2.20	-2.20
LightGBM+NWP	CTWLSV	-1.67	-1.67	-1.62	-1.62	-1.69	-1.69
LightGBM+NWP	CTBU	-1.68	-1.68	-1.61	-1.61	-1.70	-1.70
ETS (author)	CTWLSV	-0.18	-0.18	-0.22	-0.22	-0.18	-0.18
ETS (author)	CTBU	-0.17	-0.17	-0.20	-0.20	-0.17	-0.17
ETS	CTWLSV	0.83	0.83	0.74	0.74	0.85	0.85
ETS	CTBU	0.93	0.93	0.84	0.84	0.94	0.94

Table 3: Mean Skill score (%) relative to PERS by base method, reconciliation approach, spatial level, and temporal resolution.

Base method	Reconciliation	L0 H	L0 D	L1 H	L1 D	L2 H	L2 D
PERS	—	0.00	0.00	0.00	0.00	0.00	0.00
SARIMAX	CTWLSV	19.70	45.84	21.78	39.02	15.63	28.66
SARIMAX	CTBU	17.91	45.61	20.51	38.92	14.99	28.59
SARIMAX+NWP	CTWLSV	34.75	48.61	29.83	40.54	19.64	29.67
SARIMAX+NWP	CTBU	34.14	48.69	29.23	40.64	19.40	29.73
RF	CTWLSV	30.34	32.86	27.71	29.68	24.39	26.21
RF	CTBU	30.18	32.76	27.60	29.62	24.33	26.16
RF+NWP	CTWLSV	31.27	33.79	28.39	30.55	23.93	27.24
RF+NWP	CTBU	30.22	33.69	27.50	30.51	23.48	27.21
LightGBM	CTWLSV	33.86	37.83	29.46	32.84	20.29	24.38
LightGBM	CTBU	33.55	37.65	29.28	32.74	20.22	24.34
LightGBM+NWP	CTWLSV	33.11	37.16	28.91	32.74	19.60	25.29
LightGBM+NWP	CTBU	30.90	36.56	27.33	32.36	18.90	25.14
ETS (author)	CTWLSV	33.93	34.38	31.54	31.94	26.84	28.34
ETS (author)	CTBU	34.16	35.17	31.59	32.63	26.82	28.68
ETS	CTWLSV	34.02	34.96	31.70	32.59	27.05	29.24
ETS	CTBU	34.27	35.85	31.77	33.40	27.04	29.65

Table 4: Mean SMAPE (%) by base method, reconciliation approach, spatial level, and temporal resolution.

Base method	Reconciliation	L0 H	L0 D	L1 H	L1 D	L2 H	L2 D
PERS	—	10.85	7.74	12.94	9.77	20.20	13.05
SARIMAX	CTWLSV	57.06	4.70	57.91	6.53	61.21	9.86
SARIMAX	CTBU	57.28	4.74	58.18	6.56	61.31	9.88
SARIMAX+NWP	CTWLSV	55.91	4.34	57.12	6.23	60.89	9.60
SARIMAX+NWP	CTBU	56.24	4.33	57.57	6.21	61.08	9.58
RF	CTWLSV	53.63	5.46	55.30	7.17	59.09	9.89
RF	CTBU	53.67	5.47	55.32	7.18	59.09	9.90
RF+NWP	CTWLSV	54.50	5.34	55.94	7.05	59.84	9.71
RF+NWP	CTBU	55.21	5.35	56.73	7.06	60.11	9.72
LightGBM	CTWLSV	54.94	5.16	56.42	6.97	60.52	10.20
LightGBM	CTBU	54.98	5.18	56.46	6.98	60.52	10.20
LightGBM+NWP	CTWLSV	55.46	5.20	57.07	6.97	61.16	10.07
LightGBM+NWP	CTBU	56.07	5.25	57.57	7.00	61.35	10.09
ETS (author)	CTWLSV	55.66	5.22	56.96	6.91	60.28	9.54
ETS (author)	CTBU	56.00	5.16	57.40	6.85	60.46	9.49
ETS	CTWLSV	55.58	5.12	56.89	6.78	60.18	9.35
ETS	CTBU	55.96	5.05	57.38	6.70	60.39	9.29

5.4 Limitations

[TBD: Single dataset, deterministic forecasts, replication constraints.]

6 Conclusion

6.1 Summary

[TBD: Answer to the research question. Best-performing base methods per hierarchy level.]

6.2 Outlook

[TBD: Additional base methods, probabilistic forecasts, other datasets, alternative reconciliation approaches.]

References

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Appendices

A Reconciliation Implementation

[TBD: CTWLSV and CTBU details, FoReco format transformation. See repository docs/RECONCILIATION_METHODS.md.]

B Replication Bug Fixes

[TBD: File loading order and FoReco format mismatch. See docs/BUG_FIXES.md.]

C Supplementary Results

[TBD: Optional intermediate temporal levels, example forecast plot.]

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